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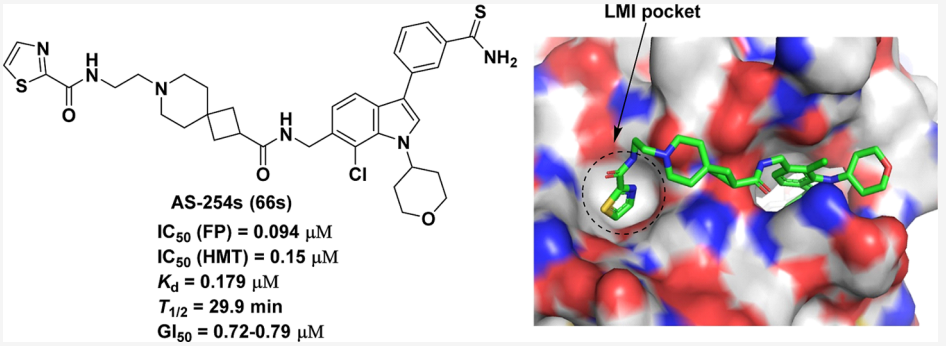
Structure-Based Development of Novel Spiro-Piperidine ASH1L Inhibitors

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ABSTRACT: The absent, small, or homeotic-like 1 (ASH1L) protein is a histone lysine methyltransferase that plays a crucial role in various cancers, including leukemia. Despite representing an attractive therapeutic target, only one class of ASH1L inhibitors was identified to date. Herein, we report development of advanced ASH1L inhibitors targeting the catalytic SET domain, which were designed to access previously unexplored binding pocket on ASH1L. Extensive medicinal chemistry combined with structure-based design led to identification of **66s** (**AS-254s**), a highly potent and selective ASH1L inhibitor (IC_{50} = 94 nM), representing substantially improved inhibitory activity over previously reported compounds targeting ASH1L. Furthermore, **66s** effectively blocked cell proliferation and induced apoptosis and differentiation in leukemia cells harboring *MLL1* translocations. Overall, this work provides a high-quality chemical probe targeting the catalytic SET domain of ASH1L with increased inhibitory activity and cellular efficacy to study biological functions of ASH1L and potentially to develop novel anticancer therapeutics.

INTRODUCTION

Modifications of chromatin, including histone lysine methylation, play essential roles in physiological and disease states.[1,2] The ASH1L (absent, small, or homeotic-like 1) protein belongs to the family of histone lysine methyltransferases, which catalyzes the mono- and demethylation of histone 3 lysine 36 (H3K36), representing an activating chromatin mark.[3−6] ASH1L is a large protein (>3000 amino acids) that contains multiple structural domains, including a catalytic SET domain and three chromatin reader domains (bromodomain, PHD and BAH domains) in addition to a long unstructured region at the N-terminal portion of the protein.[3] The SET domain of ASH1L catalyzes the transfer of a methyl group from an *S*-adenosyl methionine (SAM) cofactor to the lysine substrate (H3K36).[3,7]

ASH1L was shown to regulate the expression of *HOXA* genes and plays an important role in development of acute leukemia with translocations of the mixed lineage leukemia 1 (*MLL1*) gene (also known as *KMT2A*).[4,8−11] *MLL1* translocations are found in ∼5−10% of acute leukemia patients, leading to a poor clinical outcome with only 35% 5 year survival,[12−14] highlighting the need for new therapeutics. *ASH1L* knockdown was shown to induce cell growth arrest, apoptosis, and differentiation, associated with downregulation of *HOXA9* genes in leukemia cells with *MLL1* translocations, and it also abrogates development of leukemia in mice.[8] We have previously shown the importance of the catalytic SET domain of ASH1L in leukemogenesis mediated by *MLL1* fusion proteins,[15] supporting that small molecule inhibitors of the ASH1L SET domain could result in new antileukemic agents. Furthermore, ASH1L overexpression was found in different solid tumors, including thyroid, breast, gastric and liver cancers, and was linked to cancer cell growth and aggressive disease.[16−20] All these findings

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underscore ASH1L as an attractive therapeutic target in hematologic and solid tumors.

We have recently published the first-in-class small molecule inhibitors targeting the catalytic SET domain of ASH1L,[15] which constitute the only ASH1L inhibitors reported to date. These compounds, represented by **AS-85** (compound **1**) and **AS-99** (compound **2**), bind to the SET domain of ASH1L with submicromolar binding affinities and selectively inhibit its catalytic activity with the IC_{50} values of 0.6 and 0.79 μM, respectively (Figure 1A), without blocking other histone

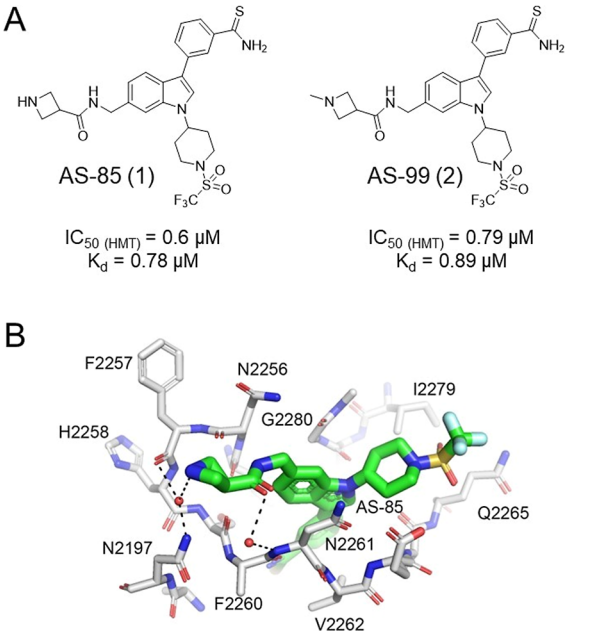


Figure 1. Known ASH1L inhibitors. (A) Chemical structures and activities of known ASH1L inhibitors. IC_{50} values were measured using HMT assay and K_d values were assessed using ITC.[15] (B) Binding mode of **AS-85** to ASH1L based on the crystal structure of the complex (6WZW in PDB). Color coding: carbon atoms are in gray (for ASH1L) or green (for **AS-85**), nitrogen atoms are in blue, oxygens in red, sulfur in yellow and fluorine atoms in light blue.

methyltransferases (HMTs).[15] **AS-99** also demonstrated antileukemic activity in leukemic cells with *MLL1* translocations, reinforcing that blocking the catalytic SET domain of ASH1L leads to antileukemic effects. Here, we present the structure-based development of new ASH1L inhibitors derived from **AS-85** and **AS-99** to occupy an additional pocket on the SET domain of ASH1L. Additionally, we developed a fluorescence polarization (FP) assay to effectively assess the activity of new ASH1L inhibitors and to determine a systematic structure−activity relationship (SAR) for these compounds. The most potent ASH1L inhibitor we developed, **66s** (**AS-254s**), demonstrated much more pronounced in vitro inhibitory activity and stronger antileukemic effect in blocking cell proliferation, inducing apoptosis and differentiation in *MLL1*-rearranged leukemia cells when compared to the previous ASH1L inhibitors.[15] Overall, this work provides new valuable chemical probes targeting the ASH1L SET domain with substantially improved inhibitory activity and cellular efficacy to study biological functions of ASH1L and to potentially develop new anticancer therapeutics.

RESULTS AND DISCUSSION

Development of an FP Assay to Assess ASH1L Inhibitors. The inhibitory activity of our first-in-class ASH1L inhibitors (e.g., **AS-85** and **AS-99**, Figure 1A) was assessed using the histone methyltransferase (HMT) assay.[15] The HMT assay is laborious, expensive, and requires radioactive SAM as a cofactor as well as a nucleosome substrate. To efficiently characterize the activity of ASH1L inhibitors, we developed an FP assay, utilizing a fluorescein-labeled ASH1L ligand. Based on the crystal structure of ASH1L with bound **AS-85** (Figure 1B), we replaced the azetidine moiety with an aliphatic linker harboring terminal fluorescein, resulting in compound **3** (Figure 2A), representing the FP probe. We also introduced a 7-chloro

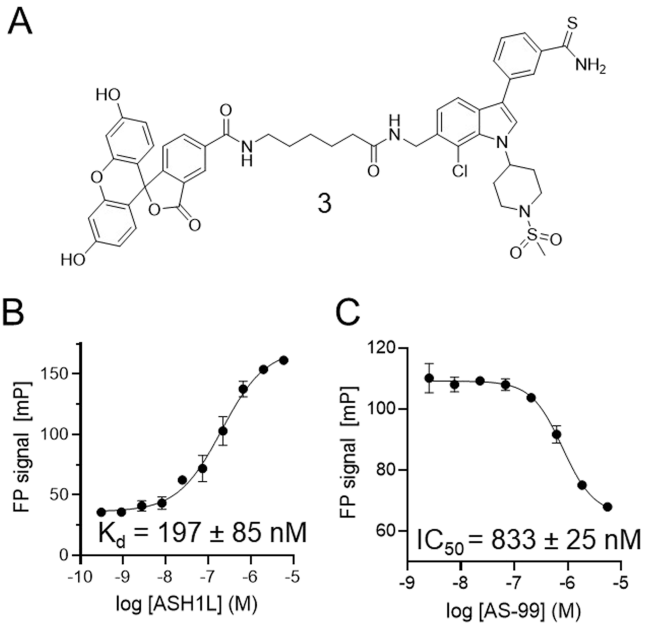


Figure 2. Development of FP assay for ASH1L. (A) Chemical structure of fluorescein-labeled probe **3**. (B). Titration curve for K_d determination for binding of **3** to ASH1L SET using FP assay. (C) Titration curve and IC_{50} value for **AS-99** from FP assay with ASH1L SET.

substituent at the indole ring of **3** to form additional interactions with the backbone amide of G2280 (Figure 1B) for further potency improvement of the FP probe. In addition, we utilized a methyl sulfonamide moiety in compound **3** over the trifluoromethyl group present in **AS-85** (Figure 2A) to improve aqueous solubility of the FP probe. Subsequently, we assessed the binding affinity (K_d) of the fluorescein-labeled compound **3** to ASH1L by FP, resulting in K_d = 197 nM and high dynamic range (~120 mP), Figure 2B. We then evaluated this new FP assay in competition experiments with our earlier ASH1L inhibitors **AS-99** and **AS-6**,[15] resulting in the IC_{50} values of 833 and 439 nM, respectively (Figures 2C and S1). Importantly, inhibitory activities obtained for these compounds in the FP assay are very similar to the IC_{50} values from the HMT assay (IC_{50} = 790 and 520 nM for **AS-99** and **AS-6**, respectively),[15] Figures 1A and S1. These results validate our new FP assay for testing novel ASH1L inhibitors reported here.

Structure-Based Design of New ASH1L Inhibitors to Occupy Unexplored Hydrophobic Pocket. We previously reported first-in-class ASH1L inhibitors, including **AS-85** and **AS-99**, which inhibit ASH1L enzymatic activity at submicro-

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